ANSWER KEY

Lab 7 Brute force catch

Capture the traffic in Wireshark (Windows 11 PC, sign in with the password networkwalks) and follow the solution steps below.

Task 1. Capture the traffic — 10 marks

Sign in to the Windows PC and capture the network traffic. How many packets did Wireshark capture?

Answer: 369

Solution: Sign in with the password networkwalks, double-click Wireshark on the desktop, then double-click the WiFi interface. The status bar shows Packets: 369.

Task 2. The protocol — 10 marks

The server owner sees many failed sign-ins. Which protocol are they on? Check Statistics › Protocol Hierarchy.

Answer: FTP

Solution: Open Statistics › Protocol Hierarchy. Under TCP you find FTP. FTP uses the commands USER and PASS.

Task 3. The target — 10 marks

Which IP address is the FTP server that was attacked?

Answer: 192.168.10.20

Solution: Filter

```
ftp
```

. Packets that say Response: 530 Login incorrect come from 192.168.10.20. That is the server.

Task 4. The attacker — 10 marks

Which IP address sent all the login attempts?

Answer: 192.168.10.66

Solution: Filter

```
ftp.request.command == "PASS"
```

. Every packet comes from 192.168.10.66.

Task 5. The username — 10 marks

Which username did the attacker try again and again?

Answer: admin

Solution: Filter

```
ftp.request.command == "USER"
```

. Every line says Request: USER admin.

Task 6. Failure code — 10 marks

Which FTP reply code means “Login incorrect”?

Answer: 530

Solution: Filter

```
ftp.response.code >= 500
```

. The Info column shows Response: 530 Login incorrect.

Task 7. Failed logins — 10 marks

How many failed logins did the attacker make?

Answer: 24

Solution: Filter

```
ftp.response.code == 530
```

. The status bar shows Displayed: 24.

Task 8. Success code — 10 marks

One login finally worked. Which FTP reply code shows a successful login?

Answer: 230

Solution: Filter

```
ftp.response.code == 230
```

. One packet shows Response: 230 Login successful.

Task 9. The password that worked — 10 marks

What password did the attacker use in the successful login?

Answer: Sunshine99

Solution: Filter

```
ftp.response.code == 230
```

, right-click the packet and choose Follow › TCP Stream. The red line PASS Sunshine99 comes just before 230 Login successful.

Task 10. The attack type — 10 marks

Many password guesses for one user, then one success. What type of attack is this?

Answer: Brute force

Solution: Trying many passwords for one user until one works is a brute force attack.

SUMMARY

Attacker:	192.168.10.66
Target:	FTP server 192.168.10.20
Failed logins:	24 (code 530)
Worked:	admin / Sunshine99 (code 230)
Attack type:	Brute force

Watch the Wireshark video playlist: https://www.youtube.com/playlist?list=PLG-wUGWU39ro